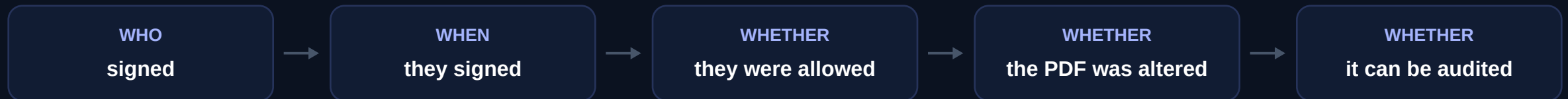


Digital Signature Workflow for University OD & Permission Documents

A secure, auditable workflow from request creation to verified final PDF

PDF preparation · multiple institutional signers · sequential approvals · signature collection · integrity · auditability

THE KEY PROBLEM IS NOT SIMPLY "GETTING A SIGNATURE". THE SYSTEM MUST ENSURE:



THREE KILLER TESTS · VERIFIED

KT1

Sequential Signing

Each signer can act only when it is their turn.

✓ PASS

KT2

PDF Tamper Detection

A single modified byte in the final PDF is detected.

✓ PASS

KT3

Complete Audit Trail

Every event is recorded as signed, chained evidence.

✓ PASS

From Request to Verified Document



HOW IT IS ENFORCED

- ✓ PostgreSQL is the source of truth
- ✓ Server-side authorization controls every workflow mutation
- ✓ Signing order is enforced server-side
- ✓ Final PDF bytes are hashed with SHA-256
- ✓ Integrity evidence is persisted in a signed manifest
- ✓ Audit events are chained and cryptographically signed
- ✓ Final document retrieval verifies integrity before serving the PDF

Three Killer Tests — Where the Rebuild Proves Its Value

KT1

SEQUENTIAL SIGNING



- ✓ Advisor cannot sign before Student
- ✓ HoD cannot sign before Advisor
- ✓ Server-side checks enforce the current signer
- ✓ Concurrent signing attempts cannot bypass the order

RESULT

✓ PASS

KT2

PDF TAMPER DETECTION

INTEGRITY CHAIN

- Final PDF
- SHA-256 digest
- Integrity manifest
- Retrieval verification

TAMPER TEST

- Modify 1 byte
- Hash mismatch
- **INTEGRITY_FAILED**
- **PDF NOT SERVED**
- Failure audited

RESULT

✓ PASS

KT3

COMPLETE AUDIT TRAIL

EVENTS RECORDED

- Creation
- Signer assignment
- Signing
- Completion
- Sealing
- PDF view
- Integrity verification / failure

EACH EVENT CARRIES

actor	document
timestamp	sequence
hash chain	cryptographic signature

RESULT

✓ PASS

Beyond Basic Digital Signing

IMPROVEMENT 1

Stronger Integrity & Evidence

A complete final-document integrity lifecycle

- ✓ Immutable final PDF version
- ✓ SHA-256 hash of the exact stored bytes
- ✓ Signed integrity manifest
- ✓ Verification before final PDF retrieval
- ✓ Explicit INTEGRITY_FAILED state
- ✓ Evidence preserved after tampering
- ✓ Append-only audit evidence

Integrity/evidence design — not certificate-backed legal PDF signing.

IMPROVEMENT 2

Configurable Completeness Assistant

Before submission, checks configured example rules for potentially missing information and gives explainable guidance to the requester.

✓ Purpose

✓ Dates / date range

✓ Destination / event

✓ Supporting information

Draft request



Rule check



Guidance



Submit

Configurable example rules — not official university policy.

"Instead of only collecting signatures, the rebuild helps ensure that the request is complete, traceable and verifiable."

Verified Results on a Rebuild-First Foundation

VERIFIED RESULTS

KT1 Sequential signing

✓ PASS

KT2 PDF tamper detection

✓ PASS

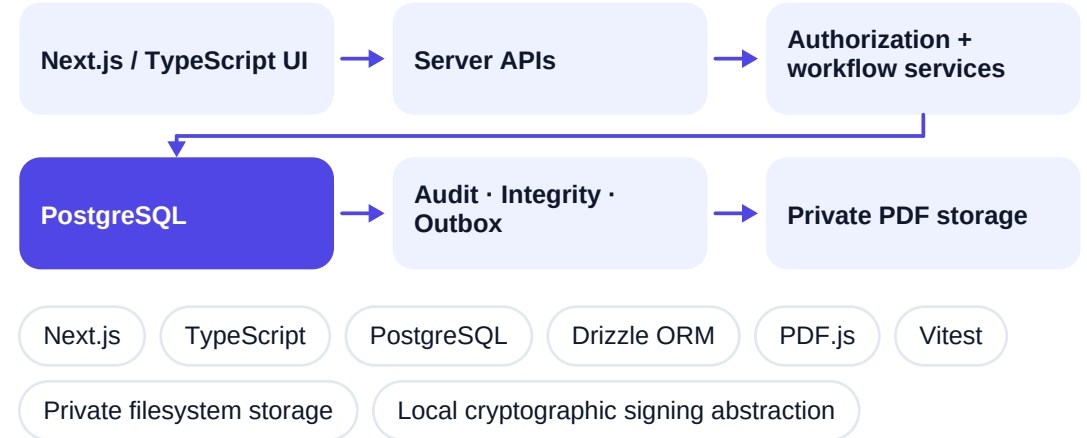
KT3 Complete audit trail

✓ PASS

ACCEPTANCE VALIDATION

- ✓ Fresh PostgreSQL validation passed
- ✓ All migrations applied successfully
- ✓ 53+ tests · full database-backed suite passed · 0 skipped
- ✓ Typecheck, lint, formatting, db:check, production build passed
- ✓ Clean-room requirement passed

ARCHITECTURE AT A GLANCE



- Clean-room rebuild — independent implementation
- Replaceable provider boundaries
- Local/demo infrastructure — no paid external dependency
- Local application identity · local/demo signing

From a basic document-signing flow to a verifiable, auditable university workflow.